

Companies working on lithium-air batteries

Only a handful of labs around the world, including those at PolyPlus Battery, Berkeley, CA, National Institute of Advanced Industrial Science and Technology (AIST), Japan, and St Andrews University, Scotland, are currently working on lithium-air batteries.

IBM. IBM is developing powerful, lightweight lithium-air batteries. It plans to make the storage capacity of the batteries ten times higher than the existing batteries. The company will partner with US National Labs to carry out the project.

National Institute of Advanced Industrial Science and Technology (AIST), Japan. National Institute of AIST began its operations on April 1, 2001. It's an amalgamation of 15 research institutes. According to them, the lithium-air battery developed has a unique structure and is most suitable for automobiles.

PolyPlus Battery. PolyPlus battery is a privately held company located in the State of California and started in the year 1990. It was founded by Dr Steve J. Visco, a principal investigator at the Lawrence Berkeley and a senior scientist at LBNL. Recently, PolyPlus Battery Company partnered with a manufacturing firm to develop the Li-air battery. It also wishes to develop rechargeable Li-air batteries within a few years.

St Andrews University. University of St Andrews is developing an air-fuelled battery, in partnership with University of Strathclyde and University of Newcastle. The research work is funded by the Engineering and Physical Sciences Research Council. This four-year research project, which began in 2007, is scheduled to end in 2011.

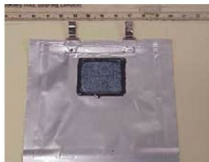


Fig. 1: Pouch cell

Arthur Doble papers and how Li-air pouch cells are used to test the air cathode.

Lithium-air batteries are constructed in layers like a sandwich. For experimental purpose, a pouch cell is used (Fig. 1), which contains an air cathode inside the Li-air cells. The cell construction further involves layering of the components and sealing of them in plastic packets. The anode, separator, electrolyte and carbon air cathodes are sealed inside the metallised plastic packets.

The anode is composed of a lithium metal foil. It is pressed into a nickel mesh current collector that was welded to a nickel tab. The area of the anode is slightly larger than the cathode, by 10 cm², to provide full usage of the air cathode. Setela, an organic polymer, is used as the separator and

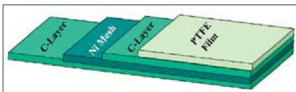


Fig. 2: Layered air cathode

the liquid electrolyte used is 1M LiPF₆ in 1:1:1 ethyle carbonate, diethyl carbonate and dimethyl carbonate. The air cathode used is a carbon composite by combining carbon, a metal catalyst and a binder deposited on a metal collector. The binder is Teflon combined with latex and the metal current collector is nickel expanded metal mesh. The air cathode structure is a layered composite with an increased capacity. A thin Teflon film between the air cathode and the atmosphere repels atmospheric water but allows oxygen diffusion.

Cathode preparation. On both sides of metal current collector, two layers of carbon, binder and catalyst mixture are deposited (refer Fig. 2). The carbon mats are impregnated with a mixture of carbon, binder and catalyst. Two of these mats are placed on either side of the current collector. The air cathodes have a microporous Teflon layer added to the side exposed to the environment, producing

double-sided air cathode.

Testing the air cathodes. The Li-air pouch cell is used to test the air cathodes. Transitional metals were added as catalysts into the carbon electrode to increase the specific capacity of the cathode. Testing results showed a specific capacity of 3471 mAh/g at 1mA discharge. The capacity of a battery depends on the anode and the cathode. Adding transitional material to it increases the specific capacity of the cathode. Some materials are better than the others. Like, if we compare manganese, silver, platinum and cobalt, manganese acts the best and increases the performance.

The future

So far so good, but what is the future of the Li-air battery which is still under development and where can it find applications? Can this battery suffice all the devices? Lot many questions like these need to be answered.

Yardney Technical Products Inc., a subsidiary of ENER-TEK International, focuses on speciality battery technology for research and product development in high-performance niche markets. Kris Johannesen, director-business development, Yardney Technical Products, says, "The future of lithium-air is continued development of components such as the anode, electrolyte, separator and especially air cathode. Cell packaging and battery design will also be developed. That will depend upon applications which include powering: cellphones, laptops, other electronic devices and even some vehicles. The recharge ability of the cell is the biggest unknown at this time. Further evaluation is necessary to optimise the performance of this promising technology. Efforts to increase the number of cycles available as well as reduce packaging size are on the roadmap to a viable Li-air system."

Moreover, Johannesen adds, some applications might require a higher